

MODULE 2 · ABLATION STUDY

Enemy-AI Evaluation Results

Module 2 — AI Believability

Per-player comparison

Player

P0001

	Basic	Intermediate	Advanced
BELIEVABILITY (SURVEY)			
Perceived Intelligence (/5)	1.4/5	4/5	4.8/5
Animacy (life-like) (/5)	1.7/5	4.3/5	4.7/5
Tactical realism (/5)	1.7/5	3.7/5	5/5
UEQ — Pragmatic (/7)	4.8/7	6/7	6.5/7
UEQ — Hedonic (/7)	6.3/7	6.5/7	7/7
OBJECTIVE (TELEMETRY)			
Enemy hit-rate (NPC) (%)	—	—	—

Overall averages (all players)			
	Basic n=20	Intermediate n=14	Advanced n=11
BELIEVABILITY (SURVEY)			
Perceived Intelligence (/5)	1.6	3.7	4.4
Animacy (life-like) (/5)	2	3.8	4.1
Tactical realism (/5)	1.7	3.5	4.4
UEQ — Pragmatic (/7)	4.6	5.3	5.7

	Basic n=20	Intermediate n=14	Advanced n=11
UEQ — Hedonic (/7)	4.7	5.4	5.8
OBJECTIVE (TELEMETRY)			
Enemy hit-rate (NPC) (%)	—	—	—

Statistical significance

Repeated-measures, non-parametric. **Friedman** asks whether the AI level changed the measure at all; **Wilcoxon** signed-rank post-hoc asks which pair differs (α is Bonferroni-corrected for the 3 pairs). A result is significant when $p < \alpha$.

Perceived Intelligence

n = 9 complete

Friedman $\chi^2(2) = 10.89$, $p = 0.004$, Kendall's $W = 0.6$ **significant**

Basic vs Intermediate	p = 0.008	r = 0.85	✓ sig
Basic vs Advanced	p = 0.008	r = 0.85	✓ sig
Intermediate vs Advanced	p = 0.016	r = 0.89	✓ sig

Post-hoc $\alpha = 0.02$ (Bonferroni · 3 pairs) · r = effect size.

Animacy (life-like)

n = 9 complete

Friedman $\chi^2(2) = 14$, $p = < 0.001$, Kendall's $W = 0.78$ **significant**

Basic vs Intermediate	p = 0.004	r = 0.89	✓ sig
Basic vs Advanced	p = 0.004	r = 0.89	✓ sig
Intermediate vs Advanced	p = 0.188	r = 0.51	n.s.

Post-hoc $\alpha = 0.02$ (Bonferroni · 3 pairs) · r = effect size.

Tactical realism

n = 9 complete

Friedman $\chi^2(2) = 11.72$, $p = 0.003$, Kendall's $W = 0.65$ **significant**

Basic vs Intermediate	p = 0.008	r = 0.89	✓ sig
Basic vs Advanced	p = 0.008	r = 0.85	✓ sig
Intermediate vs Advanced	p = 0.008	r = 0.88	✓ sig

Post-hoc $\alpha = 0.02$ (Bonferroni · 3 pairs) · $r =$ effect size.

Enemy hit-rate (NPC)

n = 0 complete

Not enough complete data for this measure (n = 0).

Survey scores are participant ratings after each play (higher = better for the AI). Enemy hit-rate is an objective telemetry measure of the same thing (how competently the AI fought). The significance section runs a Friedman test (any difference across the three levels?) and Wilcoxon signed-rank post-hoc tests (which pairs differ?) on complete-case participants — because AI difficulty IS Module 2’s research question.

Module 3 — Cognitive

Per-player average (all AI levels combined)

Reaction time doesn't depend on which AI tier was played, so each player's sessions are averaged together regardless of level.

Player	Sessions	Reaction time
PUBLISHED REFERENCE VALUES		
Expert	—	0.30 s
Novice	—	0.50 s
TRAINEES		
P0001	4	0.50 s Novice
P0002	3	1.20 s Novice
P0003	3	0.80 s Novice
P0004	3	0.90 s Novice
P0005	5	0.70 s Novice
P0006	3	0.60 s Novice
P0007	6	0.30 s Expert
P0008	3	0.70 s Novice

Player	Sessions	Reaction time
P0009	6	0.60 s Novice
P0010	5	0.70 s Novice
P0011	4	0.60 s Novice

vs Expert Value (all sessions pooled)

One-sample Wilcoxon signed-rank test: does the pooled set of every recorded session's value differ significantly from the published expert benchmark? Only measures with an independent published benchmark are testable this way.

Reaction time

n = 45 sessions

Trainee mean 0.65 s vs expert 0.30 s (Hick's Law — choice reaction time) — Wilcoxon $p = < 0.001$, $r = 0.67$ **significantly different from expert**

One-sample Wilcoxon signed-rank, $\alpha = 0.05 \cdot r = \text{effect size } (|z|/\sqrt{n})$.

ANOVA Table (between players)

One-way analysis of variance, with each PLAYER as a group and their own sessions as that group's replicates: does the mean differ between players by more than their own session-to-session variability would explain? Source / SS (sum of squares) / df (degrees of freedom) / MS (mean square) / F (F-statistic) / p — standard ANOVA-table columns.

Reaction time

k = 11 players

Source	SS	df	MS	F	p
Between players	1.94	10	0.19	1.22	0.316 n.s.
Within players	5.43	34	0.16	—	—
Total	7.37	44	—	—	—

Grand mean 0.65 s across N = 45 sessions. $F(10, 34) = 1.22$, $p = 0.316$ — no significant difference between players on reaction time. For reference, the published expert value is 0.30 s (Hick's Law — choice reaction time).

This page only surfaces reaction time as it relates to the trainee population. Module 3's full cognitive evaluation (movement telemetry, workload reasoning, stability/attention) lives in each session's own Movement and Workload tabs on the dashboard, not here.

Module 4 — Score Validity

Per-player average (all AI levels combined)

Module 4's five scores aren't about AI difficulty, so each player's sessions are averaged together across whichever levels they played.

Player	Sessions	Overall	Hostage Safety	Accuracy	Speed	Operator Safety
PUBLISHED REFERENCE VALUES						
Expert	—	—	85%	100%	—	—
Novice	—	—	40%	35%	—	—
TRAINEES						
P0001	4	50%	50% Novice	30% Novice	70%	—
P0002	3	50%	70% Proficient	30% Novice	70%	40%
P0003	3	50%	30% Novice	50% Novice	70%	—
P0004	3	80%	100% Expert	40% Novice	60%	—
P0005	5	70%	70% Proficient	70% Developing	90%	60%
P0006	3	70%	70% Proficient	100% Expert	30%	20%
P0007	6	50%	30% Novice	70% Developing	60%	30%
P0008	3	70%	50% Novice	100% Expert	60%	30%
P0009	6	70%	60% Developing	80% Proficient	60%	20%
P0010	5	50%	30% Novice	60% Developing	50%	20%
P0011	4	50%	30% Novice	100% Expert	40%	20%

vs Expert Value (all sessions pooled)

One-sample Wilcoxon signed-rank test: does the pooled set of every recorded session's value differ significantly from the published expert benchmark? Only measures with an independent published benchmark are testable this way.

Hostage Safety

n = 45 sessions

Trainee mean 52% vs expert 85% (RAND hostage-rescue outcomes) — Wilcoxon $p = < 0.001$, $r = 0.49$
significantly different from expert

One-sample Wilcoxon signed-rank, $\alpha = 0.05 \cdot r = \text{effect size } (|z|/\sqrt{n})$.

Accuracy

n = 45 sessions

Trainee mean 68% vs expert 100% (NYPD SOP-9, normalized to this session's engagement range) — Wilcoxon $p = < 0.001$, $r = 0.88$ **significantly different from expert**

One-sample Wilcoxon signed-rank, $\alpha = 0.05 \cdot r = \text{effect size } (|z|/\sqrt{n})$.

ANOVA Table (between players)

One-way analysis of variance, with each PLAYER as a group and their own sessions as that group's replicates: does the mean differ between players by more than their own session-to-session variability would explain? Source / SS (sum of squares) / df (degrees of freedom) / MS (mean square) / F (F-statistic) / p — standard ANOVA-table columns.

Overall

k = 11 players

Source	SS	df	MS	F	p
Between players	0.58	10	0.06	0.71	0.709 n.s.
Within players	2.8	34	0.08	—	—
Total	3.38	44	—	—	—

Grand mean 59% across N = 45 sessions. $F(10, 34) = 0.71$, $p = 0.709$ — no significant difference between players on overall.

Hostage Safety

k = 11 players

Source	SS	df	MS	F	p
Between players	1.88	10	0.19	0.76	0.666 n.s.
Within players	8.41	34	0.25	—	—
Total	10.29	44	—	—	—

Grand mean 52% across N = 45 sessions. $F(10, 34) = 0.76$, $p = 0.666$ — no significant difference between players on hostage safety. For reference, the published expert value is 85% (RAND hostage-rescue outcomes).

Accuracy

k = 11 players

Source	SS	df	MS	F	p
Between players	2.7	10	0.27	2.87	0.011 sig.
Within players	3.2	34	0.09	—	—
Total	5.9	44	—	—	—

Grand mean 68% across N = 45 sessions. $F(10, 34) = 2.87$, $p = 0.011$ — players differ significantly from each other on accuracy. For reference, the published expert value is 100% (NYPD SOP-9, normalized to this session's engagement range).

Speed

k = 11 players

Source	SS	df	MS	F	p
Between players	0.95	10	0.1	2.26	0.037 sig.
Within players	1.43	34	0.04	—	—
Total	2.38	44	—	—	—

Grand mean 61% across N = 45 sessions. $F(10, 34) = 2.26$, $p = 0.037$ — players differ significantly from each other on speed.

Operator Safety

k = 8 players

Source	SS	df	MS	F	p
Between players	0.64	7	0.09	4.22	0.003 sig.
Within players	0.59	27	0.02	—	—
Total	1.23	34	—	—	—

Grand mean 30% across N = 35 sessions. $F(7, 27) = 4.22$, $p = 0.003$ — players differ significantly from each other on operator safety.

Repeated-Play Reliability

Is Module 4's score consistent when the SAME person plays more than once? Shown per session, with mean and standard deviation across sessions — tight clustering (low SD) means the score is reliable, not noisy.

P0001

4 sessions

	Session 1	Session 2	Session 3	Session 4	Mean	SD
Overall	76%	84%	19%	13%	48%	±37pp
Hostage Safety	100%	100%	0%	0%	50%	±58pp
Accuracy	47%	64%	0%	0%	28%	±33pp
Speed	57%	72%	97%	65%	72%	±17pp
Operator Safety	—	—	—	—	—	—

P0002

3 sessions

	Session 1	Session 2	Session 3	Mean	SD
Overall	18%	76%	68%	54%	±31pp
Hostage Safety	0%	100%	100%	67%	±58pp
Accuracy	23%	35%	18%	25%	±9pp
Speed	57%	76%	65%	66%	±9pp
Operator Safety	20%	45%	41%	35%	±13pp

P0003

3 sessions

	Session 1	Session 2	Session 3	Mean	SD
Overall	74%	36%	27%	45%	±25pp
Hostage Safety	100%	0%	0%	33%	±58pp
Accuracy	40%	67%	33%	47%	±18pp
Speed	59%	77%	84%	73%	±13pp
Operator Safety	—	—	—	—	—

P0004

3 sessions

	Session 1	Session 2	Session 3	Mean	SD
Overall	75%	80%	72%	76%	±4pp
Hostage Safety	100%	100%	100%	100%	±0pp
Accuracy	43%	56%	29%	43%	±14pp
Speed	62%	67%	66%	65%	±2pp
Operator Safety	—	—	—	—	—

P0005

5 sessions

	Session 1	Session 2	Session 3	Session 4	Session 5	Mean	SD
Overall	81%	92%	34%	75%	92%	75%	±24pp
Hostage Safety	100%	100%	0%	61%	100%	72%	±44pp
Accuracy	44%	82%	55%	77%	83%	68%	±17pp
Speed	87%	87%	88%	86%	87%	87%	±1pp
Operator Safety	65%	58%	62%	55%	62%	60%	±4pp

P0006

3 sessions

	Session 1	Session 2	Session 3	Mean	SD
Overall	40%	82%	88%	70%	±26pp
Hostage Safety	0%	100%	100%	67%	±58pp
Accuracy	100%	100%	100%	100%	±0pp
Speed	50%	12%	38%	33%	±20pp
Operator Safety	0%	18%	28%	15%	±14pp

P0007

6 sessions

	Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	Mean	SD
Overall	20%	27%	47%	70%	90%	45%	50%	±26pp
Hostage Safety	0%	0%	0%	100%	100%	0%	33%	±52pp
Accuracy	0%	77%	100%	45%	100%	100%	70%	±41pp
Speed	100%	20%	83%	32%	52%	77%	61%	±31pp
Operator Safety	20%	11%	20%	22%	43%	61%	29%	±19pp

P0008

3 sessions

	Session 1	Session 2	Session 3	Mean	SD
Overall	90%	65%	44%	66%	±23pp
Hostage Safety	100%	37%	0%	46%	±51pp
Accuracy	100%	100%	100%	100%	±0pp
Speed	49%	49%	72%	56%	±13pp
Operator Safety	15%	15%	50%	27%	±20pp

P0009

6 sessions

	Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	Mean	SD
Overall	46%	94%	92%	15%	72%	91%	68%	±32pp
Hostage Safety	0%	100%	100%	0%	61%	100%	60%	±49pp
Accuracy	100%	100%	100%	0%	100%	100%	83%	±41pp
Speed	78%	71%	59%	77%	36%	57%	63%	±16pp

	Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	Mean	SD
Operator Safety	20%	16%	10%	20%	25%	40%	22%	±10pp

P0010

5 sessions

	Session 1	Session 2	Session 3	Session 4	Session 5	Mean	SD
Overall	7%	87%	79%	16%	43%	47%	±36pp
Hostage Safety	0%	100%	61%	0%	0%	32%	±46pp
Accuracy	22%	100%	100%	0%	100%	64%	±49pp
Speed	0%	36%	73%	82%	67%	51%	±34pp
Operator Safety	0%	18%	42%	20%	19%	20%	±15pp

P0011

4 sessions

	Session 1	Session 2	Session 3	Session 4	Mean	SD
Overall	38%	91%	32%	40%	50%	±27pp
Hostage Safety	0%	100%	0%	0%	25%	±50pp
Accuracy	100%	100%	100%	100%	100%	±0pp
Speed	39%	55%	10%	47%	38%	±20pp
Operator Safety	20%	50%	6%	23%	25%	±19pp

vs Expert Benchmarks (averaged per player)

Each person's sessions are averaged first (to smooth out single-session noise), then compared to published expert/novice values where one exists. Speed and Operator Safety have no independent published benchmark — shown as-is, not scored against a source that doesn't exist.

P0001

avg of 4 sessions

Overall

48%

No independent published benchmark exists for this composite score.

Hostage Safety

50%

NOVICE

Novice 40% · Expert 85% · RAND hostage-rescue outcomes

Accuracy

28%

NOVICE

Novice 35% · Expert 100% · NYPD SOP-9, normalized to this session's engagement range

Speed

72%

No independent published benchmark exists for this composite score.

P0002

avg of 3 sessions

Overall

54%

No independent published benchmark exists for this composite score.

Hostage Safety

67%

DEVELOPING

Novice 40% · Expert 85% · RAND hostage-rescue outcomes

Accuracy

25%

NOVICE

Novice 35% · Expert 100% · NYPD SOP-9, normalized to this session's engagement range

Speed

66%

No independent published benchmark exists for this composite score.

Operator Safety

35%

No independent published benchmark exists for this composite score.

P0003

avg of 3 sessions

Overall

45%

No independent published benchmark exists for this composite score.

Hostage Safety

33%

NOVICE

Novice 40% · Expert 85% · RAND hostage-rescue outcomes

Accuracy

47%

NOVICE

Novice 35% · Expert 100% · NYPD SOP-9, normalized to this session's engagement range

Speed

73%

localhost:3000/evaluation

12/16

No independent published benchmark exists for this composite score.

P0004

avg of 3 sessions

Overall

76%

No independent published benchmark exists for this composite score.

Hostage Safety

100%

EXPERT

Novice 40% · Expert 85% · RAND hostage-rescue outcomes

Accuracy

43%

NOVICE

Novice 35% · Expert 100% · NYPD SOP-9, normalized to this session's engagement range

Speed

65%

No independent published benchmark exists for this composite score.

P0005

avg of 5 sessions

Overall

75%

No independent published benchmark exists for this composite score.

Hostage Safety

72%

PROFICIENT

Novice 40% · Expert 85% · RAND hostage-rescue outcomes

Accuracy

68%

DEVELOPING

Novice 35% · Expert 100% · NYPD SOP-9, normalized to this session's engagement range

Speed

87%

No independent published benchmark exists for this composite score.

Operator Safety

60%

No independent published benchmark exists for this composite score.

P0006

avg of 3 sessions

Overall

70%

No independent published benchmark exists for this composite score.

Hostage Safety

67%

DEVELOPING

Novice 40% · Expert 85% · RAND hostage-rescue outcomes

Accuracy

100%

EXPERT

Novice 35% · Expert 100% · NYPD SOP-9, normalized to this session's engagement range

Speed

33%

No independent published benchmark exists for this composite score.

Operator Safety

15%

No independent published benchmark exists for this composite score.

P0007

avg of 6 sessions

Overall

50%

No independent published benchmark exists for this composite score.

Hostage Safety

33%

NOVICE

Novice 40% · Expert 85% · RAND hostage-rescue outcomes

Accuracy

70%

DEVELOPING

Novice 35% · Expert 100% · NYPD SOP-9, normalized to this session's engagement range

Speed

61%

No independent published benchmark exists for this composite score.

Operator Safety

29%

No independent published benchmark exists for this composite score.

P0008

avg of 3 sessions

Overall

66%

No independent published benchmark exists for this composite score.

Hostage Safety

46%

NOVICE

Novice 40% · Expert 85% · RAND hostage-rescue outcomes

Accuracy

100%

EXPERT

Novice 35% · Expert 100% · NYPD SOP-9, normalized to this session's engagement range

Speed

56%

No independent published benchmark exists for this composite score.

Operator Safety

27%

No independent published benchmark exists for this composite score.

P0009

avg of 6 sessions

Overall

68%

No independent published benchmark exists for this composite score.

Hostage Safety

60%

DEVELOPING

Novice 40% · Expert 85% · RAND hostage-rescue outcomes

Accuracy

83%

PROFICIENT

Novice 35% · Expert 100% · NYPD SOP-9, normalized to this session's engagement range

Speed

63%

No independent published benchmark exists for this composite score.

Operator Safety

22%

No independent published benchmark exists for this composite score.

P0010

avg of 5 sessions

Overall

47%

No independent published benchmark exists for this composite score.

Hostage Safety

32%

NOVICE

Novice 40% · Expert 85% · RAND hostage-rescue outcomes

Accuracy

64%

DEVELOPING

Novice 35% · Expert 100% · NYPD SOP-9, normalized to this session's engagement range

Speed

51%

No independent published benchmark exists for this composite score.

Operator Safety

20%

No independent published benchmark exists for this composite score.

P0011

avg of 4 sessions

Overall

50%

No independent published benchmark exists for this composite score.

Hostage Safety

25%

NOVICE

Novice 40% · Expert 85% · RAND hostage-rescue outcomes

Accuracy

100% EXPERT

Novice 35% · Expert 100% · NYPD SOP-9, normalized to this session's engagement range

Speed

38%

No independent published benchmark exists for this composite score.

Operator Safety

25%

No independent published benchmark exists for this composite score.